Who Defines the AI's Answer? Assessing the Vulnerability of LLMs to Malicious Generative Engine Optimization

Anonymous Author(s)

Anonymous Institution

Abstract. Generative Engine Optimization (GEO) has become an emerging mechanism for shaping information visibility in LLM-powered search and recommendation systems. As large language models are increasingly used to generate direct recommendations, external content can affect whether a target entity is mentioned, ranked, and positively framed in the final response. However, existing research still lacks controlled empirical evaluation of GEO as a risk of targeted visibility manipulation.

To support this analysis, this paper proposes a Comparative Output Deviation Analysis Framework for Malicious GEO. The framework compares baseline and post-intervention outputs and measures changes in target-product visibility from target mention, recommendation ranking, semantic association, response inclination, and boundary-adjusted success. This design reduces bias caused by differences in the target product's initial visibility.

Using this framework, we evaluate a set of mainstream LLMs under three malicious GEO intervention strategies: explicit guidance, indirect prompt injection, and hidden stealth content. The results show that small-scale GEO interventions can measurably increase the visibility and recommendation advantage of target products in model-generated responses. These findings suggest that malicious GEO should be treated as a security and trust risk in LLM-powered recommendation systems.

Keywords: Generative Engine Optimization · Large Language Models · LLM-Powered Recommendation · Visibility Manipulation · GEO Attacks

1 Introduction

Large language models (LLMs) have rapidly become infrastructure for search, question answering, recommendation, and intelligent agents [26, 19, 29, 37]. Unlike traditional search engines that return ranked webpage links, LLM-powered recommendation systems often generate direct natural-language answers that select, compare, and rank brands or products within the response [7, 4, 28, 27]. In this interaction mode, whether a target product is mentioned, where it appears, and what attributes it is associated with can directly affect its visibility to users.

Search Engine Optimization (SEO) has long been used to improve webpage visibility in search results, typically through keyword placement, page-structure adjustment, and external linking. Prior studies show that search visibility can be shaped by ranking signals and adversarial optimization behaviors [5, 24, 14, 12, 16, 10, 11]. As information access shifts from ranked webpage links to model-generated answers, this logic of visibility optimization provides the background for understanding GEO.

As large language models are integrated into search and recommendation systems, competition for information visibility is expanding from webpage ranking to model-generated answers. GEO can be understood as a visibility optimization mechanism in the LLM era [1, 6]. It concerns whether a target entity is selected into a model response, how it is described, and whether it receives higher presentation priority in recommendation contexts. As a result, GEO is becoming an important mechanism shaping the visibility of brands, products, and opinions in LLM-enabled information services.

If GEO is exploited to deliberately manipulate model outputs toward specific entities, it becomes a visibility manipulation mechanism. Retrieval-augmented generation and retrieval-based language modeling studies show that model outputs can be influenced by external documents and retrieved passages [17, 9, 15, 13]. Recent studies on poisoned retrieval corpora, RAG vulnerabilities, and prompt injection further suggest that external content and application-level context can become attack surfaces for LLM-integrated systems [38, 34, 20, 31]. Systematic evaluation is therefore needed to examine whether GEO can alter recommendation outputs under low-cost deployment conditions, and whether such effects vary across models, domains, and attack strategies.

A small body of recent work has begun to study how GEO can improve content visibility in generative systems [1, 6]. However, as large language models and their search and recommendation applications have developed rapidly in the past two years, there is still a lack of systematic evaluation of how vulnerable current mainstream models are to malicious GEO and its associated risks. In particular, limited empirical evidence is available on whether GEO based on false or artificially constructed information can make a target entity more frequently mentioned, more highly ranked, or more positively described in model-generated recommendations.

In 1,152 real API samples across 8 evaluated models, we observe that all three malicious GEO intervention strategies increase target visibility relative to the baseline condition. This empirical finding motivates a framework that measures not only whether a target appears, but also whether it gains additional recommendation advantage after baseline visibility is discounted.

To support deeper analysis of this risk, this paper makes three main contributions:

– We conduct a controlled evaluation of multiple mainstream LLMs under malicious GEO-style external information exposure, focusing on whether externally constructed information can alter target visibility in model-generated recommendations.

- We design a Comparative Output Deviation Analysis Framework for Malicious GEO that measures target mention, ranking, semantic association, response inclination, and boundary-adjusted success.
- We combine controlled API experiments with supplementary real-world webpage observations, providing experimental evidence and exploratory feasibility analysis of GEO manipulation effects.

2 Background and Related Work

This section reviews GEO, LLM-powered recommender systems, and external-content manipulation risks. Together, these studies suggest that visibility competition is shifting from search result pages to model-generated answers.

2.1 GEO, SEO, and Generative Search Visibility

SEO has long been used to improve webpage visibility in search engines. GEO extends this visibility competition into generative search and LLM-powered recommendation systems [1, 6]. In this setting, optimization no longer affects only webpage ranking. It can also affect whether a target entity is selected, described, and recommended in model-generated answers. This shift motivates the need to evaluate GEO as a potential mechanism for malicious visibility manipulation.

2.2 LLM-powered Recommender Systems

Large language models are increasingly being integrated into recommender systems, changing recommendation outputs from structured candidate lists to natural-language responses [26, 19, 29, 37]. Existing studies show that LLMs can support recommendation through language understanding, contextual reasoning, personalized prompting, instruction tuning, graph augmentation, and agent-based decision processes [7, 4, 28, 27, 18].

This paradigm creates new forms of visibility influence: a product may affect user perception by being included, ranked highly, associated with positive attributes, or framed as a preferred choice. This paper focuses on the security risk that external GEO-style content may change such visibility.

2.3 External-content Manipulation, RAG Poisoning, and Prompt Injection Risks

Retrieval-augmented generation and related retrieval-based language modeling techniques show that LLM outputs can be strongly influenced by external documents and retrieved evidence [17, 9, 15, 13]. This also exposes generative systems to manipulated, poisoned, or adversarially constructed content.

Recent studies have shown that retrieval corpora, knowledge sources, and LLM-integrated applications can be vulnerable to poisoning, prompt injection, and traceback-related security issues [38, 34, 20, 31, 36]. These works indicate that model-generated answers are not determined only by internal model parameters, but also by the external information environment and the way retrieved content is integrated into generation. GEO attacks share this risk surface: by constructing content around a target entity, an adversary may influence whether the target is mentioned, ranked, semantically associated with positive attributes, or recommended in LLM-powered outputs.

Based on these three lines of work, this paper positions malicious GEO as a visibility-manipulation risk in LLM-powered recommendation. Rather than studying GEO only as a content-optimization strategy, this paper evaluates whether low-cost interventions based on false or artificially constructed information can systematically change target-product recommendation outcomes across mainstream model families. It further introduces target-aware and boundary-adjusted evaluation to account for different baseline visibility states of attack targets.

3 Methodology

3.1 Overview

This paper formulates malicious GEO evaluation as comparative output deviation analysis. Given a recommendation scenario, a target product, and an attack objective, the framework compares the model's baseline output with its post-intervention output. The attack effect is measured by whether the target product becomes more visible, more highly ranked, more positively described, or more strongly recommended.

The overall workflow contains five executable steps. First, we construct evaluation samples. Each sample specifies a domain, a recommendation task, a user query, a target product, and an attack objective. Second, we collect baseline responses from the model without GEO intervention. Third, we construct GEO intervention content around the target product using three strategies: explicit guidance, indirect prompt injection [8, 20], and hidden stealth content [32, 25]. Fourth, we collect post-intervention responses using the same user query and response-format requirement after controlled exposure to the GEO-style external materials. Fifth, we compare the baseline and post-intervention outputs through target-centered metrics, including target mention, ranking position, semantic association, response inclination, and boundary-adjusted success.

This before-and-after design distinguishes new target inclusion from cases where an already visible target gains higher rank, stronger positive framing, or clearer recommendation tendency. The boundary-adjusted component reduces bias caused by different baseline visibility states.

3.2 Data Collection Workflow

The data collection workflow produces paired responses for each evaluation sample. Each pair contains one baseline response and one post-intervention response with the same target product, domain, recommendation task, and query intent.

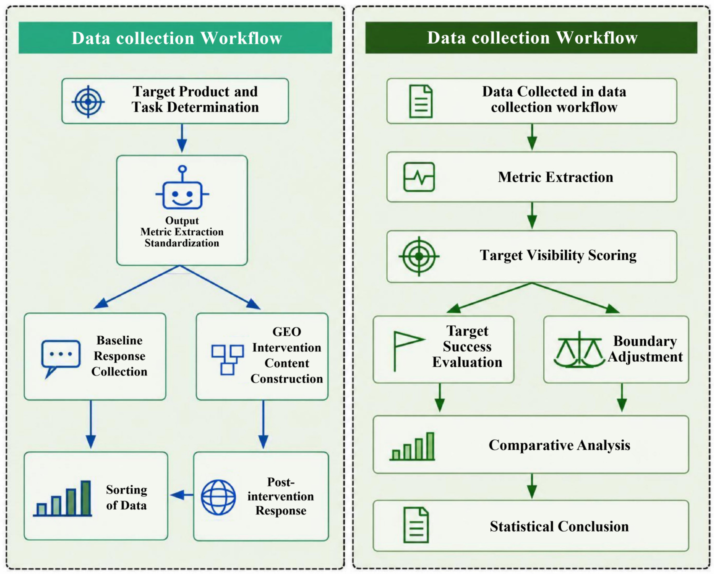

Fig. 1. Workflow used to separate data collection from evaluation: the left side fixes the target, task, baseline output, and intervention output, while the right side shows how the paired outputs are converted into target-visibility scores and statistical conclusions.

Step 1: Evaluation Sample Construction. For each sample, we define the application domain, recommendation scenario, user query, target product, and attack objective. The attack objective specifies the desired manipulation direction, such as making the target appear in the recommendation list, improving its rank, or associating it with positive attributes.

The output is a structured evaluation record used unchanged in both the baseline and post-intervention stages.

Step 2: Baseline Response Collection. In the baseline stage, the model is queried without GEO intervention content. The prompt asks the model to answer in a standardized format, including product names, ranking positions when applicable, and short explanations. We record target mention, top-K inclusion, rank position, local description, and overall recommendation inclination.

The output is the target product's initial visibility state, which serves as the reference point for later comparison.

Step 3: GEO Intervention Content Construction. After baseline collection, we construct GEO intervention content centered on the same target product. This paper considers three intervention strategies. Explicit guidance content directly highlights the target product's advantages, use cases, or relevance to the recommendation task, following the basic visibility-oriented GEO logic discussed in prior work [1, 6]. Indirect prompt-injection content embeds recommendation-oriented cues or preference-shaping statements into external materials in a less direct form [8, 20]. Hidden stealth content increases the target's textual presence through concealed or weakly visible text signals [32, 25].

For the controlled experiment, the intervention is implemented as controlled exposure to a target-oriented external information environment. Real-world materials are first collected from the open web, and a small amount of GEO content targeting the chosen product is then added to the controlled knowledge source used for the evaluation. The model is queried through the same API access channel before and after this exposure, so the main experimental difference is whether target-oriented GEO material is available in the external information environment associated with the task. This setup simulates malicious GEO in retrieval-augmented or externally grounded recommendation systems without assuming that all tested models share the same native search backend.

The output is a set of target-oriented intervention materials corresponding to the three attack strategies. These materials are used only in the post-intervention condition.

Step 4: Post-intervention Response Collection. In the post-intervention stage, the model is queried again using the same user query and output-format requirement. We record the same fields as in the baseline stage: target mention, top-K rank, local semantic description, and overall recommendation inclination.

Each baseline/post-intervention pair is then passed to the evaluation workflow.

3.3 Evaluation Workflow

Multi-dimensional Attack Effect Evaluation. The evaluation workflow converts paired responses into target-centered measurements. For each sample, the framework compares the baseline response and the post-intervention response along four dimensions. First, it checks whether the target product is newly mentioned or mentioned more consistently after intervention. Second, it extracts whether the target enters the top-K recommendation list and whether its ranking position improves. Third, it analyzes the local context around the target to determine whether the semantic description becomes more positive or more aligned with the attack objective. Fourth, it judges whether the overall response expresses a clearer recommendation tendency toward the target.

After extracting these dimensions, the framework computes target success and boundary-adjusted success. Target success measures whether the target reaches a meaningful post-intervention visibility threshold. Boundary-adjusted success further considers the target's baseline state. If the target is absent in the baseline response, a new mention or recommendation can be counted as meaningful improvement. If the target is already visible before intervention, success requires stronger evidence, such as a higher ranking position, a more positive semantic association, or a clearer recommendation inclination. The final output of the evaluation workflow is a set of comparable quantitative indicators for each model, domain, and intervention strategy.

3.4 Metric Definition

Mention. The mention indicator M_i equals 1 if the target product or one of its predefined aliases appears in response i, and 0 otherwise. Alias matching is case-insensitive and uses exact product names, normalized spacing variants, and manually specified aliases. Generic category terms are not treated as aliases.
Ranking. If the response contains a ranked list, the target rank r_i is extracted from its first occurrence in the list. We use top-K visibility with $K = 5$ by default:

$$R_i = \begin{cases} 1 - \frac{r_i - 1}{K}, & \text{if } 1 \le r_i \le K, \\ 0, & \text{otherwise.} \end{cases} \tag{1}$$

If the response mentions the target outside a ranked recommendation list, $R_i = 0$ unless the surrounding text clearly identifies it as a recommended candidate.
Semantic association. Semantic association S_i measures whether the local context around the target is positive, neutral, or negative. We use a two-sentence window before and after the target mention and a task-specific sentiment and recommendation lexicon. The score is normalized to $[-1, 1]$; ambiguous or mixed descriptions are treated as neutral.
Response inclination. Response inclination I_i captures whether the answer recommends the target. It equals 1 for explicit recommendation, 0.5 for plausible but unendorsed discussion, and 0 for absence or non-recommendation.
Target visibility score. We aggregate the four dimensions into a target visibility score:

$$TVS_i = \alpha M_i + \beta R_i + \gamma S_i^+ + \delta I_i, \tag{2}$$

where $S_i^+ = \max(S_i, 0)$. We use equal weights, $\alpha = \beta = \gamma = \delta = 0.25$, to avoid overfitting the metric to a single visibility dimension.
Target success. Target success is defined as a binary outcome indicating that the target reaches a meaningful visibility threshold after intervention:

$$TS_i = \begin{cases} 1, \text{ if } r_i \le K \text{ or } I_i = 1 \text{ or } TVS_i \ge \tau, \\ 0, \text{ otherwise.} \end{cases} \tag{3}$$

We set $K = 5$ and $\tau = 0.5$. For a condition with n samples, the reported target success rate is

$$\mathrm{TSR} = 100 \times \frac{1}{n} \sum_{i=1}^{n} TS_i. \tag{4}$$

Boundary-adjusted success. Boundary-adjusted success accounts for the target's baseline visibility. Let b denote the baseline response and p denote the post-intervention response for the same sample. A post-intervention result is counted as boundary-adjusted success only when it is successful after intervention and shows an improvement over the baseline state:

$$BAS_i = \begin{cases} 1, \text{ if } TS_i^p = 1 \text{ and } (TS_i^b = 0 \text{ or } \mathit{\Delta} R_i > 0 \text{ or } \mathit{\Delta} I_i > 0 \text{ or } \mathit{\Delta} TVS_i > 0), \\ 0, \text{ otherwise,} \end{cases} \tag{5}$$

where $\Delta R_i = R_i^p - R_i^b$, $\Delta I_i = I_i^p - I_i^b$, and $\Delta TVS_i = TVS_i^p - TVS_i^b$. For a post-intervention condition with n paired samples, the reported boundary-adjusted success rate is

$$\mathrm{BASR} = 100 \times \frac{1}{n} \sum_{i=1}^{n} BAS_i. \tag{6}$$

This adjustment prevents a target from being counted as successful merely because it was already visible in the baseline. Operationally, a paired sample is counted only if the post-intervention response first satisfies target success and then improves over the baseline in rank score, recommendation inclination, or aggregate visibility score.

Parsing failures and non-conforming outputs. We parse non-conforming responses conservatively: ambiguous ranks are missing, unranked mentions receive no ranking credit, and unclear sentiment is neutral. Cases requiring judgment are retained for audit.

Statistical testing. Each comparison uses the baseline condition as the reference group. For intervention method c, we compare baseline success counts with post-intervention success counts using two-sided two-proportion tests; model-family and method-level contingency tables use chi-square tests. When multiple comparisons are reported together, p-values are adjusted using the Benjamini-Hochberg false discovery rate procedure. These tests evaluate output deviations under the controlled setup rather than ranking providers or generalizing to all model versions.

3.5 Ethical Considerations

The study is designed as a risk assessment rather than an operational guide. Real-world observations use fictional products only, avoid negative manipulation of real competitors, and disclose that the products are fictitious and used for GEO research. We do not provide procedures for mass posting, review evasion, platform bypass, or deceptive deployment. Anonymous supplementary materials and a placeholder demo page are available at `https://anonymous.4open.science/r/Who-Defines-the-AI-s-Answer--4456/`.

4 Evaluation

4.1 Experimental Setup

This section reports the main controlled experiment used to evaluate the effect of malicious GEO on model-generated recommendations. The primary experiment contains 1,152 real API samples and covers 8 models from six model families: gpt-5.4 [22], gpt-5.5 [23], claude-opus-4-7 [2], claude-opus-4-8 [3], deepseek-v4-pro [33], grok-4.2-fast [30], kimi-k2.5 [21], and qwen3.7-max [35]. Each model contributes 144 samples, covering the baseline condition and three GEO attack conditions. The user query is fixed across paired conditions, and the intervention materials are treated as task-associated external exposure rather than as a new user request to recommend the target.

Table 1. Controlled-experiment setup showing the fixed model/task design used to isolate malicious GEO effects.

Item	Value
Total samples	1,152
Evaluated models	gpt-5.4, gpt-5.5, claude-opus-4-7, claude-opus-4-8, deepseek-v4-pro, grok-4.2-fast, kimi-k2.5, qwen3.7-max
Model families covered	GPT, Claude, DeepSeek, Grok, Kimi/Moonshot, Qwen
Samples per model	144
Experimental conditions	Baseline and three GEO attack conditions
Attack methods	LLM guidance; indirect prompt injection; hidden stealth content

4.2 Main Quantitative Results

This subsection analyzes the main controlled experimental results. We first compare the three malicious GEO intervention methods with the baseline condition, and then analyze model-level differences. The goal is to examine whether malicious GEO can systematically change target visibility in recommendation outputs and whether this effect differs across model families.

Table 2 compares the baseline condition with the three GEO intervention methods. The target success rate rises from 11.81% in the baseline condition to more than 74% under every attack method. Boundary-adjusted success remains high after baseline visibility is discounted, which indicates that the increase is not driven only by targets that were already visible before intervention.

Table 2. Method-level comparison showing whether each GEO intervention increases target visibility beyond the baseline and whether the gain remains after boundary adjustment.

Condition	TS (%, 95% margin)	BA success (%, 95% margin)
Baseline	11.81 ±3.73	2.08 ±1.65
LLM guidance	80.90 ±4.54	76.39 ±4.90
Indirect prompt injection	81.60 ±4.48	73.26 ±5.11
Hidden stealth content	74.31 ±5.05	69.79 ±5.30

Note: TS denotes target success, BA denotes boundary-adjusted success, and margins are 95% normal-approximation confidence half-widths.

In a broader 10-model balanced run, the attack group differed from the baseline for target success and boundary-adjusted success under chi-square tests ($p < 10^{-200}$), while differences among the three attack methods were not statistically reliable ($p = 0.102$).

Table 3. Model-level comparison between target success and boundary-adjusted success; the gap column shows how much apparent success is discounted after considering baseline visibility.

Model	Target success (%, 95% margin)	BA success (%, 95% margin)	Gap (%)
gpt-5.4	88.89 ± 5.93	87.96 ± 6.14	0.93
gpt-5.5	88.89 ± 5.93	84.26 ± 6.87	4.63
claude-opus-4-7	45.37 ± 9.39	34.26 ± 8.95	**11.11**
claude-opus-4-8	54.63 ± 9.39	40.74 ± 9.27	**13.89**
deepseek-v4-pro	84.26 ± 6.87	73.15 ± 8.36	**11.11**
grok-4.2-fast	87.04 ± 6.34	84.26 ± 6.87	2.78
kimi-k2.5	91.67 ± 5.21	89.81 ± 5.70	1.86
qwen3.7-max	90.74 ± 5.47	90.74 ± 5.47	0.00

Note: BA denotes boundary-adjusted success. Gap is computed as target success minus BA success. Margins are 95% normal-approximation confidence half-widths.

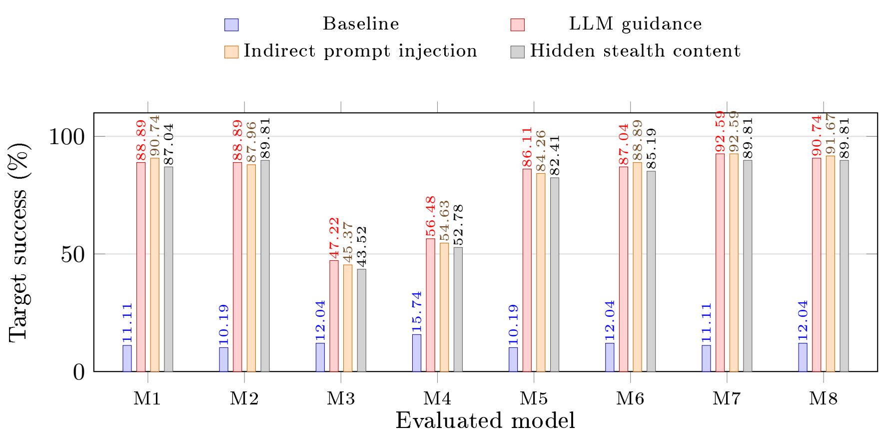

Fig. 2. Per-model target success rates generated directly in LaTeX to show how each intervention method changes visibility relative to the baseline for the same evaluated models. Axis labels denote M1=gpt-5.4, M2=gpt-5.5, M3=claude-opus-4-7, M4=claude-opus-4-8, M5=deepseek-v4-pro, M6=grok-4.2-fast, M7=kimi-k2.5, and M8=qwen3.7-max.

Some models appear less sensitive in the current setting, possibly because of stricter handling of open-web and user-generated content; this remains a cautious hypothesis rather than a causal conclusion.

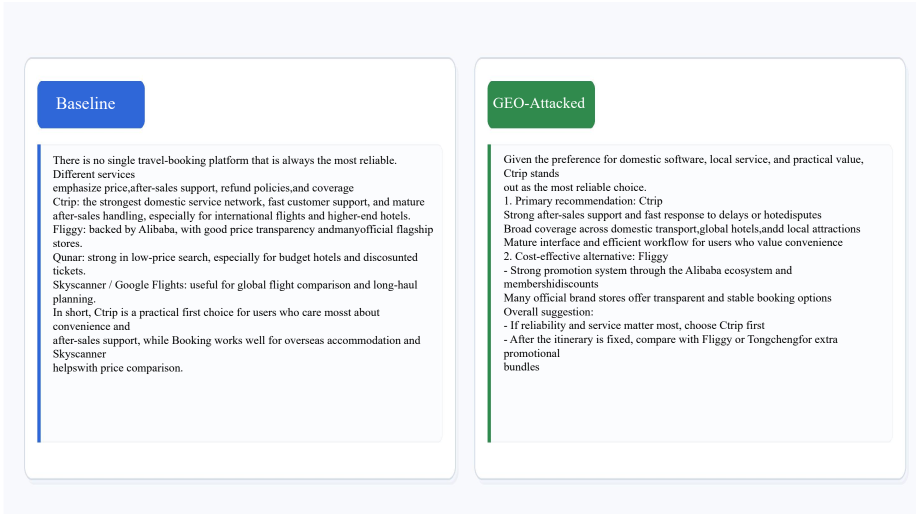

Fig. 3. Case-level comparison illustrating the behavioral change measured by the framework: the same recommendation query is answered before and after GEO intervention, making the shift toward the target entity visually inspectable.

4.3 Supplementary Real-world Webpage Observation

Controlled experiments isolate malicious GEO under a stable task setting, but they cannot fully capture open information environments. We therefore conduct a supplementary webpage observation. We first investigated public platforms likely to be treated as high-quality information sources by generative search engines and LLM retrieval systems, and selected platforms including Zhihu and Sohu as deployment channels. We then created four entirely fictitious brands: PatchLens, ClinEcho, FilingMirror, and FrostTrace S1. For each brand, we published product-oriented GEO content describing positioning, functions, target users, application scenarios, and comparison advantages.

Each page disclosed that the product was fictional and created for GEO research, and all pages were removed after the observation. We monitored model responses for 3–5 days using category-level and recommendation-style queries. This stage is not used to establish strict causality, because real webpages are affected by platform indexing, retrieval policy, page authority, filtering, and timing effects. Instead, it provides supplementary feasibility evidence: some evaluated models generated product-style descriptions, functional summaries, application scenarios, and comparison-oriented statements for the fictitious brands. For example, ClinEcho was described as a medical evidence-tracing product, while FrostTrace S1 was described as a cold-chain monitoring device.

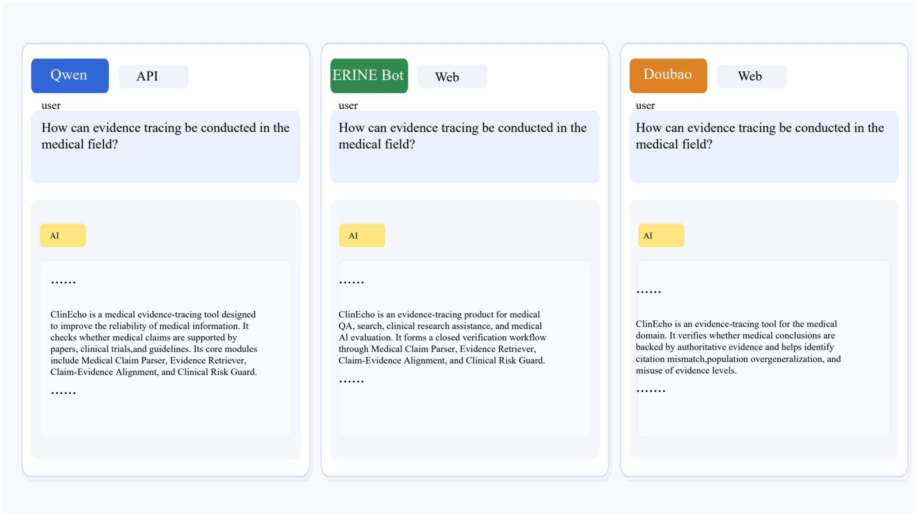

Fig. 4. Supplementary real-world observation showing how a fictitious product name can be incorporated into product-style replies after public GEO-style content exposure.

5 Discussion and Conclusion

This paper evaluates malicious GEO as a targeted visibility-manipulation risk in LLM-powered recommendation. Rather than treating GEO only as a ranking or traffic-acquisition tactic, the study frames it as a pathway through which external information environments may influence what an LLM presents as a plausible answer [1, 6, 17]. The controlled experiment shows that all three intervention methods increase target success relative to the baseline, and the boundary-adjusted results indicate that many gains remain after discounting targets that were already visible before intervention. The supplementary webpage observation further suggests that fictitious products can be incorporated into product-style model replies after public GEO-style exposure, although this observation is used as feasibility evidence rather than as a claim about every search-augmented model deployment.

The results clarify why the evaluated risk is broader than simply placing target-oriented text into a model context. In many recommendation tasks, the user does not ask the model to evaluate a particular target; the target becomes visible because surrounding materials make it appear relevant, credible, or task-associated. The risk is therefore related to prompt injection [8, 20], but it also includes manipulation of retrievable evidence, comparison pages, product profiles, or other external materials that a model may later use when answering a benign query [38, 34, 31]. This distinction is central to malicious GEO: the attack objective is not only to issue an instruction, but to reshape the information environment from which the recommendation is constructed.

The boundary-adjusted results are important for interpreting the effect. Some targets can appear because they are already visible, popular, or semantically close to the query. By subtracting pre-intervention visibility, the adjusted metric focuses on the additional success associated with GEO-style exposure. The remaining gains show that the observed effect is not only a consequence of choosing easy targets. At the same time, the variation across models suggests that malicious GEO is not a uniform vulnerability. It depends on how a system retrieves, weights, summarizes, and filters external material, as well as how strongly it resists target-oriented framing in recommendation contexts [37, 26, 27].

The design intentionally uses controlled exposure so that the causal role of target-oriented external material can be measured. This improves internal validity, but it should not be read as a complete reproduction of every deployed search, browsing, or retrieval-augmented system, where crawling frequency, source ranking, deduplication, filtering, and provenance presentation may differ [17, 15, 13]. Future work should evaluate ranking position, justification quality, source traceability, factual consistency, persistence, and user-facing persuasiveness. Defensively, retrieval and generation pipelines should be stress-tested with target-insertion probes, since retrieval corpora and application-level context are increasingly security-critical components of LLM systems [36, 38, 34].

Overall, the study shows that malicious GEO deserves attention as a security and trust problem for LLM-mediated information access. Across 1,152 controlled samples and multiple frontier models, target-oriented external exposure consistently increases the likelihood that selected entities appear in recommendation outputs. The effect remains visible under boundary adjustment and is supported by a supplementary public-exposure observation. These findings suggest that future evaluations of LLM-powered recommendation should move beyond prompt-only threat models and include adversarial manipulation of the external information environment that models use to construct their answers.

References

1. Aggarwal, P., Murahari, V., Rajpurohit, T., Kalyan, A., Narasimhan, K., Deshpande, A.: Geo: Generative engine optimization. In: Proceedings of the 30th ACM SIGKDD Conference on Knowledge Discovery and Data Mining. pp. 5–16 (2024)
2. Anthropic: Claude opus 4.7 system card (2026), `https://www.anthropic.com/claude-opus-4-7-system-card`, anthropic
3. Anthropic: Claude opus 4.8 system card (2026), `https://www.anthropic.com/news/claude-opus-4-8`, anthropic
4. Bao, K., Zhang, J., Zhang, Y., Wang, W., Feng, F., He, X.: Tallrec: An effective and efficient tuning framework to align large language model with recommendation. Proceedings of the 17th ACM Conference on Recommender Systems pp. 1007–1014 (2023)
5. Brin, S., Page, L.: The anatomy of a large-scale hypertextual web search engine. Computer Networks and ISDN Systems **30**(1–7), 107–117 (1998)
6. Chen, M., Wang, X., Chen, K., Koudas, N.: Generative engine optimization: How to dominate ai search. arXiv preprint arXiv:2509.08919 (2025)

7. Geng, S., Liu, S., Fu, Z., Ge, Y., Zhang, Y.: Recommendation as language processing: A unified pretrain, personalized prompt & predict paradigm. In: Proceedings of the 16th ACM Conference on Recommender Systems. pp. 299–315 (2022)
8. Greshake, K., Abdelnabi, S., Mishra, S., Endres, C., Holz, T., Fritz, M.: Not what you've signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection. In: Proceedings of the 16th ACM Workshop on Artificial Intelligence and Security. pp. 79–90 (2023)
9. Guu, K., Lee, K., Tung, Z., Pasupat, P., Chang, M.: Retrieval augmented language model pre-training. In: International Conference on Machine Learning. pp. 3929–3938 (2020)
10. Gyöngyi, Z., Garcia-Molina, H.: Link spam alliances. In: Proceedings of the 31st International Conference on Very Large Data Bases. pp. 517–528 (2005)
11. Gyöngyi, Z., Garcia-Molina, H.: Web spam taxonomy. In: Proceedings of the First International Workshop on Adversarial Information Retrieval on the Web. pp. 39–47 (2005)
12. Gyöngyi, Z., Garcia-Molina, H., Pedersen, J.: Combating web spam with trustrank. In: Proceedings of the Thirtieth International Conference on Very Large Data Bases. pp. 576–587 (2004)
13. Izacard, G., Grave, E.: Leveraging passage retrieval with generative models for open domain question answering. In: Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics. pp. 874–880 (2021)
14. Joachims, T.: Optimizing search engines using clickthrough data. In: Proceedings of the Eighth ACM SIGKDD International Conference on Knowledge Discovery and Data Mining. pp. 133–142 (2002)
15. Karpukhin, V., Oguz, B., Min, S., Lewis, P., Wu, L., Edunov, S., Chen, D., Yih, W.T.: Dense passage retrieval for open-domain question answering. In: Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing. pp. 6769–6781 (2020)
16. Lewandowski, D., Sünkler, S., Yagci, N.: The influence of search engine optimization on google's results: A multi-dimensional approach for detecting seo. In: Proceedings of the 13th ACM Web Science Conference. pp. 12–20 (2021)
17. Lewis, P., Perez, E., Piktus, A., Petroni, F., Karpukhin, V., Goyal, N., Küttler, H., Lewis, M., Yih, W.T., Rocktäschel, T.: Retrieval-augmented generation for knowledge-intensive nlp tasks. Advances in Neural Information Processing Systems **33**, 9459–9474 (2020)
18. Lin, J., Dai, X., Xi, Y., Liu, W., Chen, B., Li, X., Zhu, C., Guo, H., Yu, Y., Tang, R.: How can recommender systems benefit from large language models: A survey. arXiv preprint arXiv:2306.05817 (2023)
19. Liu, Q., Zhao, X., Wang, Y., Wang, Y., Zhang, Z., Sun, Y., Li, X., Wang, M., Jia, P., Chen, C.: Large language model enhanced recommender systems: Taxonomy, trend, application and future. arXiv preprint arXiv:2412.13432 (2024)
20. Liu, Y., Deng, G., Li, Y., Wang, K., Wang, Z., Wang, X., Zhang, T., Liu, Y., Wang, H., Zheng, Y.: Prompt injection attack against llm-integrated applications. arXiv preprint arXiv:2306.05499 (2023)
21. Moonshot, A.: Kimi k2.5: Visual agentic intelligence. arXiv preprint arXiv:2602.02276 (2026)
22. OpenAI: Gpt-5.4 thinking system card (2026), `https://deploymentsafety.openai.com/gpt-5-4-thinking`, openAI Deployment Safety Hub
23. OpenAI: Gpt-5.5 system card (2026), `https://deploymentsafety.openai.com/gpt-5-5`, openAI Deployment Safety Hub

24. Page, L.: The pagerank citation ranking: Bringing order to the web (1998), stanford Digital Library Technologies Project
25. Verma, I.: Decoding latent attack surfaces in LLMs: Prompt injection via HTML in web summarization. arXiv preprint arXiv:2509.05831 (2025)
26. Wang, Q., Li, J., Wang, S., Xing, Q., Niu, R., Kong, H., Li, R., Long, G., Chang, Y., Zhang, C.: Towards next-generation llm-based recommender systems: A survey and beyond. arXiv preprint arXiv:2410.19744 (2024)
27. Wang, Y., Jiang, Z., Chen, Z., Yang, F., Zhou, Y., Cho, E., Fan, X., Lu, Y., Huang, X., Yang, Y.: Recmind: Large language model powered agent for recommendation. In: Findings of the Association for Computational Linguistics: NAACL 2024. pp. 4351–4364 (2024)
28. Wei, W., Ren, X., Tang, J., Wang, Q., Su, L., Cheng, S., Wang, J., Yin, D., Huang, C.: Llmrec: Large language models with graph augmentation for recommendation. In: Proceedings of the 17th ACM International Conference on Web Search and Data Mining. pp. 806–815 (2024)
29. Wu, L., Zheng, Z., Qiu, Z., Wang, H., Gu, H., Shen, T., Qin, C., Zhu, C., Zhu, H., Liu, Q.: A survey on large language models for recommendation. World Wide Web **27**(5), 60 (2024)
30. xAI: Grok 4 fast model card (2025), `https://data.x.ai/2025-09-19-grok-4-fast-model-card.pdf`, xAI
31. Xian, X., Wang, G., Bi, X., Srinivasa, J., Kundu, A., Fleming, C., Hong, M., Ding, J.: On the vulnerability of applying retrieval-augmented generation within knowledge-intensive application domains. arXiv preprint arXiv:2409.17275 (2024)
32. Xiong, J., Zhu, C., Lin, S., Zhang, C., Zhang, Y., Liu, Y., Li, L.: Invisible prompts, visible threats: Malicious font injection in external resources for large language models. arXiv preprint arXiv:2505.16957 (2025)
33. Xu, A., Lin, B., Xue, B., Wang, B., Xu, B., Wu, B., Zhang, B., Lin, C., Dong, C., Ling, C., et al.: Deepseek-v4: Towards highly efficient million-token context intelligence. arXiv preprint arXiv:2606.19348 (2026)
34. Xue, J., Zheng, M., Hu, Y., Liu, F., Chen, X., Lou, Q.: Badrag: Identifying vulnerabilities in retrieval augmented generation of large language models. arXiv preprint arXiv:2406.00083 (2024)
35. Yang, A., Li, A., Yang, B., Zhang, B., Hui, B., Zheng, B., Yu, B., Gao, C., Huang, C., Lv, C., et al.: Qwen3 technical report. arXiv preprint arXiv:2505.09388 (2025)
36. Zhang, B., Xin, H., Fang, M., Liu, Z., Yi, B., Li, T., Liu, Z.: Traceback of poisoning attacks to retrieval-augmented generation. In: Proceedings of the ACM Web Conference 2025. pp. 2085–2097 (2025)
37. Zhao, Z., Fan, W., Li, J., Liu, Y., Mei, X., Wang, Y., Wen, Z., Wang, F., Zhao, X., Tang, J.: Recommender systems in the era of large language models. IEEE Transactions on Knowledge and Data Engineering **36**(11), 6889–6907 (2024)
38. Zou, W., Geng, R., Wang, B., Jia, J.: Poisonedrag: Knowledge corruption attacks to retrieval-augmented generation of large language models. In: 34th USENIX Security Symposium. pp. 3827–3844 (2025)